

Corrine Sadler

D598-Analytics Programming

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Task 3: Presentation

A. Code Explanation

The Python program developed in Task 2 performs all required data analysis using the pandas and NumPy libraries. The program begins by importing the D598 dataset from an Excel file located in the same directory and verifies successful loading by previewing the data. Duplicate rows are identified to ensure data integrity.

Next, the dataset is grouped by the Business State field, and descriptive statistics are calculated for all numeric variables by state. These statistics are stored in a separate dataframe to support state-level analysis. The program then filters the dataset to identify businesses with negative debt-to-equity ratios, allowing for further evaluation of potential financial risk. To calculate the debt-to-income ratio for each company, a new dataframe is created using total long-term debt and total revenue.

The debt-to-income ratio is defined as total long-term debt divided by total revenue. To prevent division-by-zero errors and preserve data integrity, an explicit conditional check is implemented using np.where. When total revenue equals zero, the debt-to-income ratio is assigned to a NaN value; otherwise, the ratio is calculated normally.

Finally, the calculated debt-to-income ratio is concatenated back to the original dataset using column-wise concatenation, producing a final dataframe that includes all original variables

along with the newly derived ratio. All codes are executed without errors and follow Python best practices for clarity, accuracy, and robustness.

B. Customized Data Visualization

Four customized visualizations were created to support stakeholder analysis:

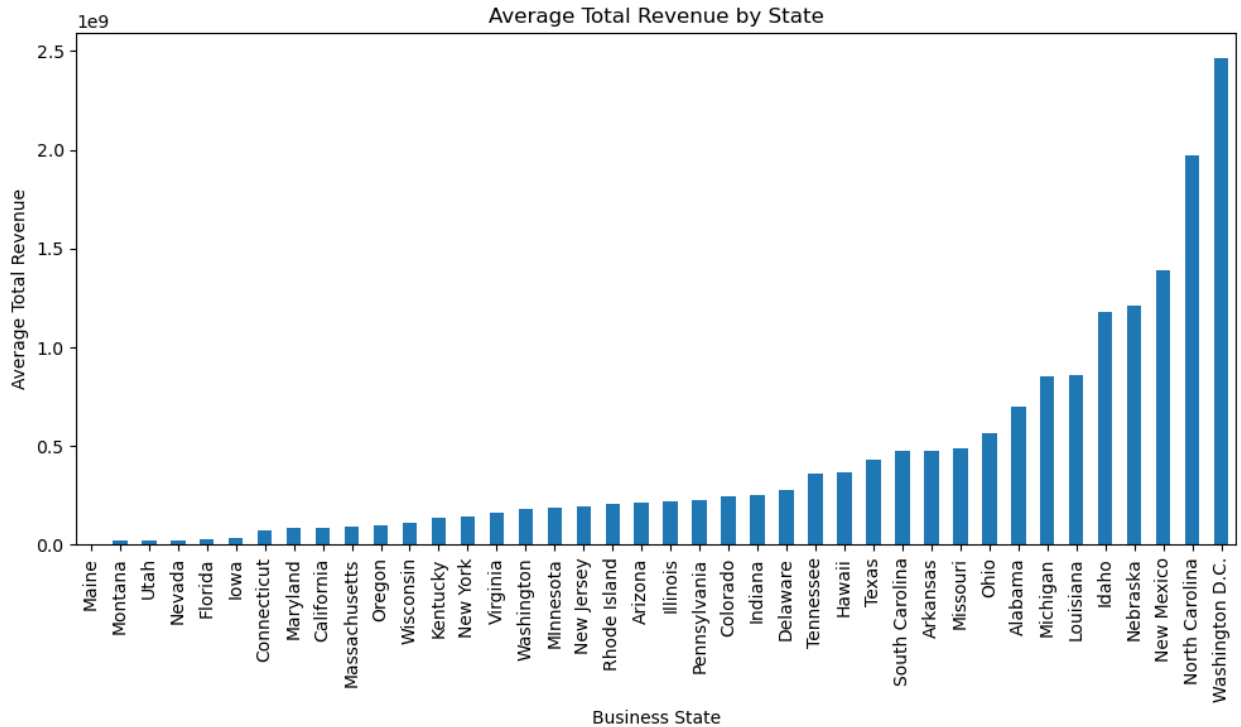


Figure 1 displays the average total revenue by state using a bar chart. This visualization allows for direct comparison of revenue performance across states with notably higher or lower average revenues.

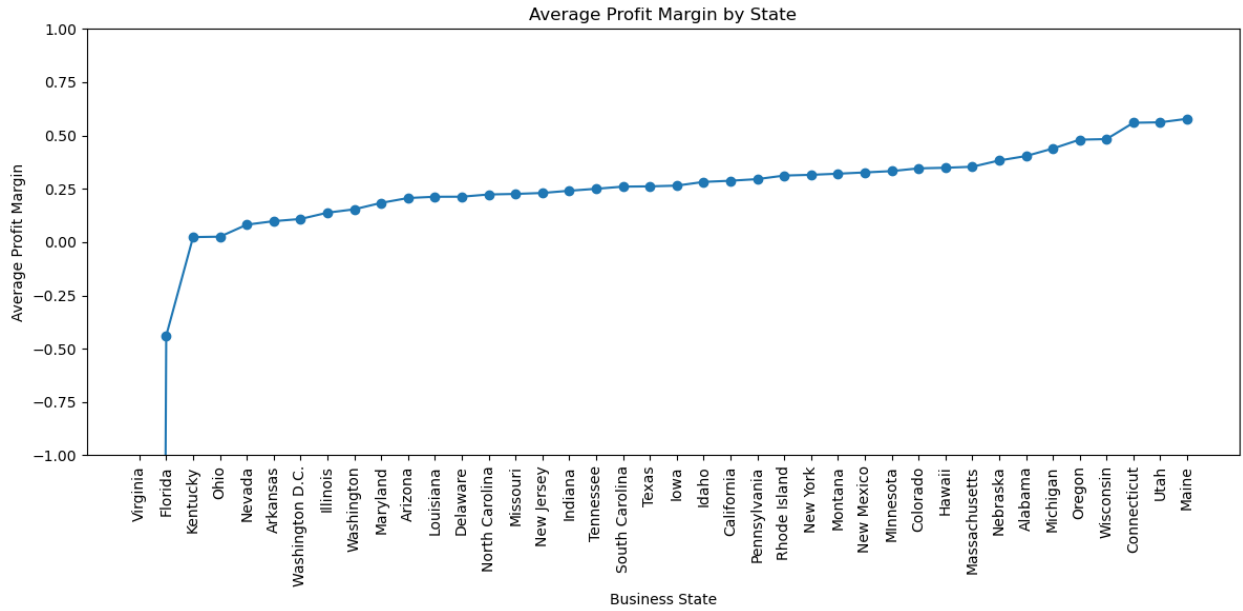


Figure 2 illustrates the average profit margin by state using a line chart. This visualization highlights variations in probability across states, making it easier to identify trends and relative differences in profit margins.

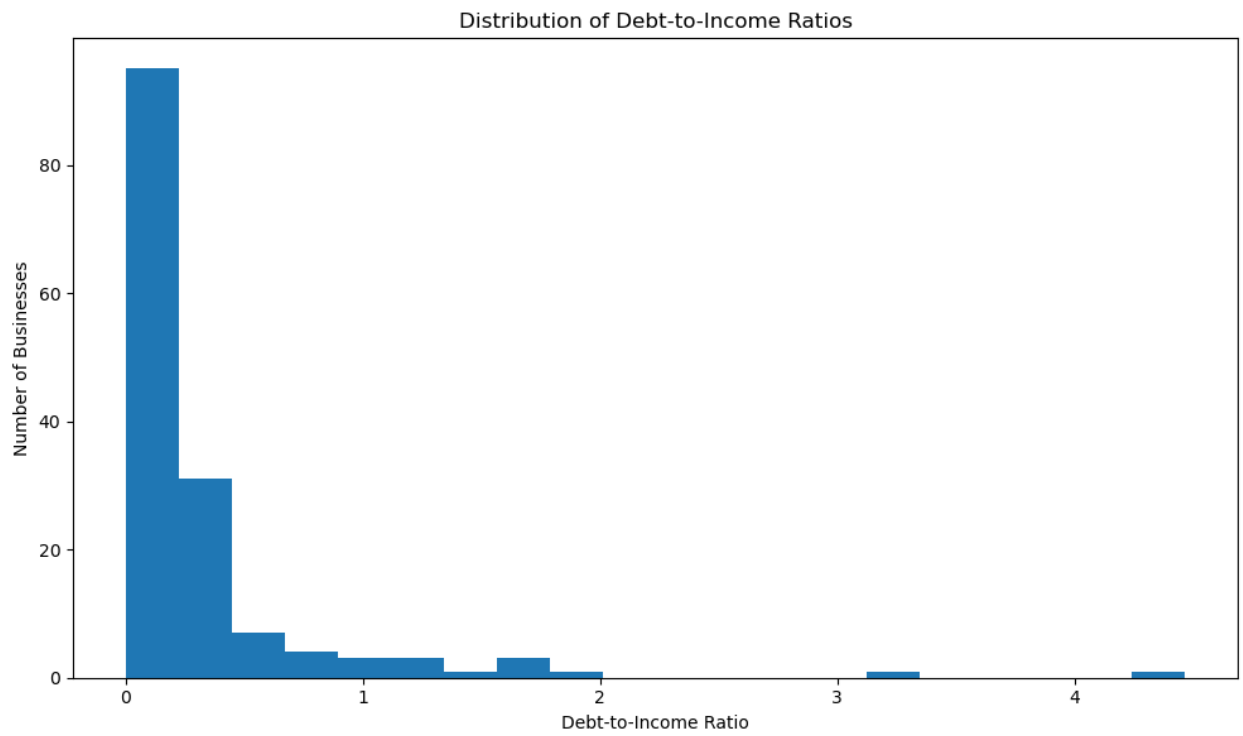


Figure 3 presents a histogram of debt-to-income ratios across all businesses. This visualization illustrates the distribution of financial leverage, highlighting both standard ratio ranges and extreme values that may indicate elevated financial risk.

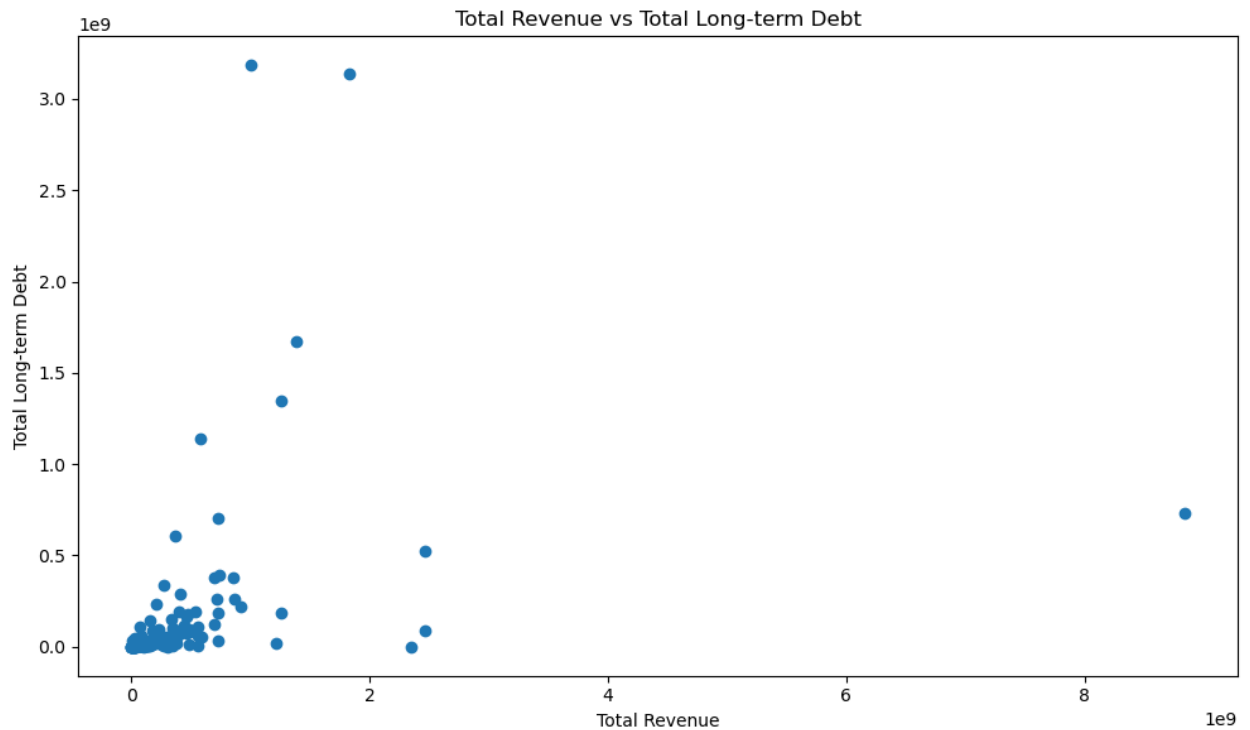


Figure 4 displays a scatter plot of total revenue versus total long-term debt. This visualization examines the relationship between revenue and debt levels, enabling the identification of clusters, trends, and outliers among businesses.

C. Explanation of Customized Visualizations

Figure 1: Average Total Revenue by State

To create this visualization, the dataset was grouped by *Business State* using the Pandas `groupby()` function. The mean of Total Revenue was calculated for each state. The resulting

values were sorted and plotted using Matplotlib as a bar chart. Axis labels, a descriptive title, and layout adjustments were added to improve readability and clarity for stakeholders.

Figure 2: Average Profit Margin by State

This visualization was created by grouping the dataset by *Business State* and calculating the average *Profit Margin* for each state. The aggregated values were sorted and plotted using a line chart in Matplotlib. Markers were added to highlight individual data points, and axis labels and rotation were applied to ensure the state name remained readable.

Figure 3: Distribution of Debt-to-Income Ratios

The debt-to-income ratio was calculated by dividing the Total Long-term Debt by Total Revenue for each business. These values were plotted using a histogram to show the overall distribution of financial leverage across companies. The histogram format allows stakeholders to quickly identify standard ranges and detect outliers that may indicate higher financial risk.

Figure 4: Total Revenue vs. Total Long-term Debt

This visualization was created using a scatter plot with Total Revenue on the x-axis and Total Long-term Debt on the y-axis. Each point represents an individual business. Matplotlib was used to plot the data, apply labels, and add a title. This visualization enables stakeholders to assess the relationship between revenue generation and debt levels across various businesses.

D. Sources

No external sources were used in this report. All analysis was conducted using the course-provided materials and the supplied dataset.